Per-site ΔRMSE from adding disturbance — M3 (C+T) vs M4 (C+T+D)

Below 0 = D reduces the site's RMSE | bars coloured by relative disturbance (site maximum) | repeated CV (3×80 %) | tree cover ≥30 % (93 sites, 372 site-years) · deadtrees v2-2

XGBoost (Optuna) · 12m | **XGBoost (Optuna) · 24m** | **LightGBM (Optuna) · 12m** | **LightGBM (Optuna) · 24m**

GPP (g C m⁻² season⁻¹)
59% improved | 66% improved | 68% improved | 61% improved
200 | 100 | 0 | -100 | -200 | -300

NEP (g C m⁻² season⁻¹)
62% improved | 60% improved | 66% improved | 65% improved
100 | 0 | -100

ET (mm season⁻¹)
49% improved | 47% improved | 49% improved | 52% improved
50 | 25 | 0 | -25

WUE (g C mm⁻¹)
60% improved | 59% improved | 58% improved | 55% improved
0.5 | 0.0 | -0.5 | -1.0

ΔRMSE = RMSE(with D) − RMSE(without D)

Sites (sorted by ΔRMSE within each panel)

Relative disturbance (%)
site maximum, 500 m
breaks 21.6 | 34.5
60
40
20
0